

## DSA Day 2 - Viva Questions & Programs

### 1. What is Traversal?

Traversal is the process of visiting every element of an array one by one.

### 2. Why do we use Traversal?

To access, update, search, display, calculate sum and average of array elements.

### 3. Two ways of Traversal

**Method 1:** for i in range(len(arr)): print(arr[i])

**Method 2:** for x in arr: print(x)

### 4. What is Accessing an Element?

Retrieving an element using its index.

### 5. What is Updating an Element?

Changing the value of an existing element.

### 6. How do you find the Sum?

Add all elements together.

### 7. How do you find the Average?

Average = Sum / Number of Elements.

## Programs (Two Methods Each)

### Traversal

Method 1:

```
arr=[10,20,30,40,50]
for i in range(len(arr)):
    print(arr[i])
```

Method 2:

```
for x in arr:
    print(x)
```

### Access Element

Method 1:

```
print(arr[2])
```

Method 2:

```
i=int(input('Enter index: '))
print(arr[i])
```

### Update Element

Method 1:

```
arr[1]=99
```

Method 2:

```
i=int(input())
v=int(input())
arr[i]=v
```

### Sum of Elements

Method 1:

```
print(sum(arr))
```

Method 2:

```
total=0
```

```
for x in arr:  
    total+=x  
print(total)
```

### **Average of Elements**

Method 1:

```
avg=sum(arr)/len(arr)
```

Method 2:

```
total=0  
for x in arr:  
    total+=x  
avg=total/len(arr)
```

### **Quick Revision**

Traversal → Visit every element

Access → arr[index]

Update → arr[index]=value

Sum → sum(arr) or loop

Average → sum(arr)/len(arr)